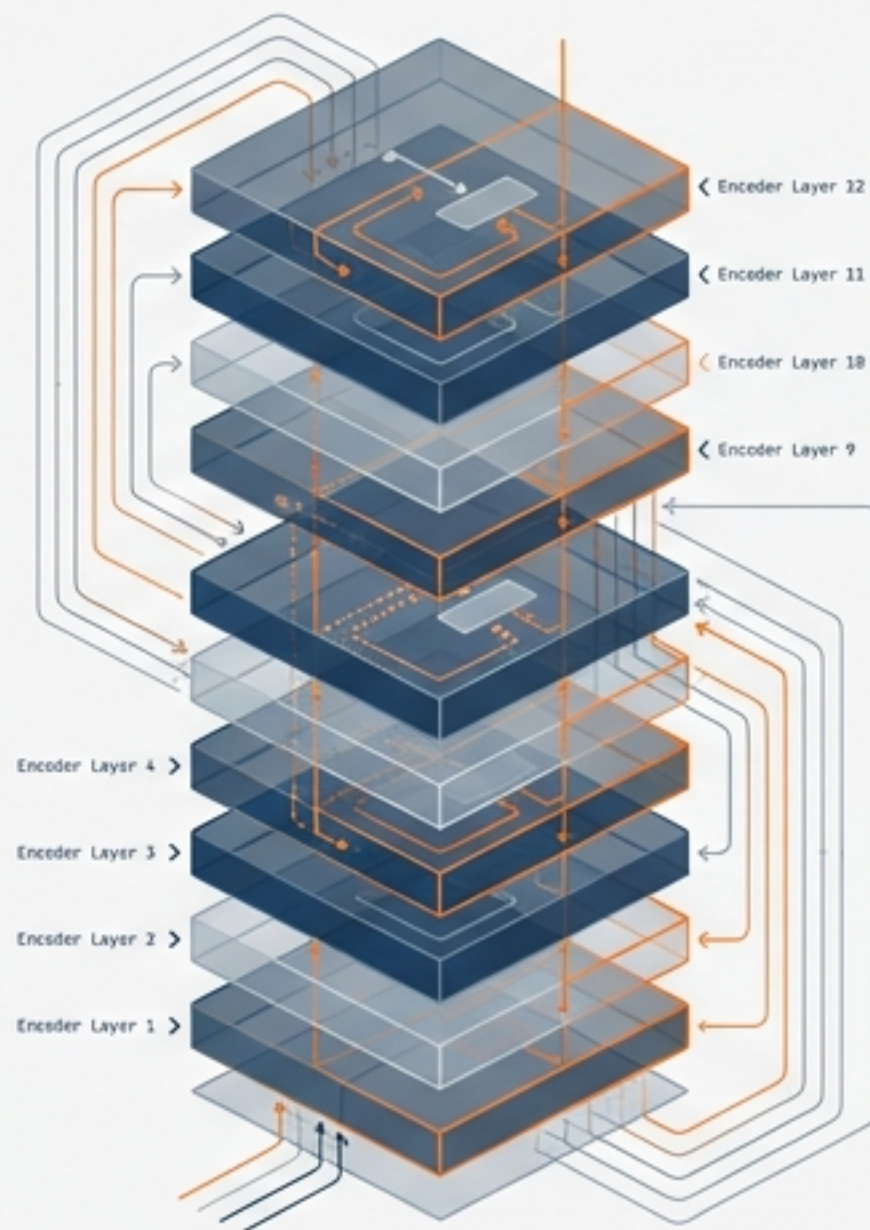
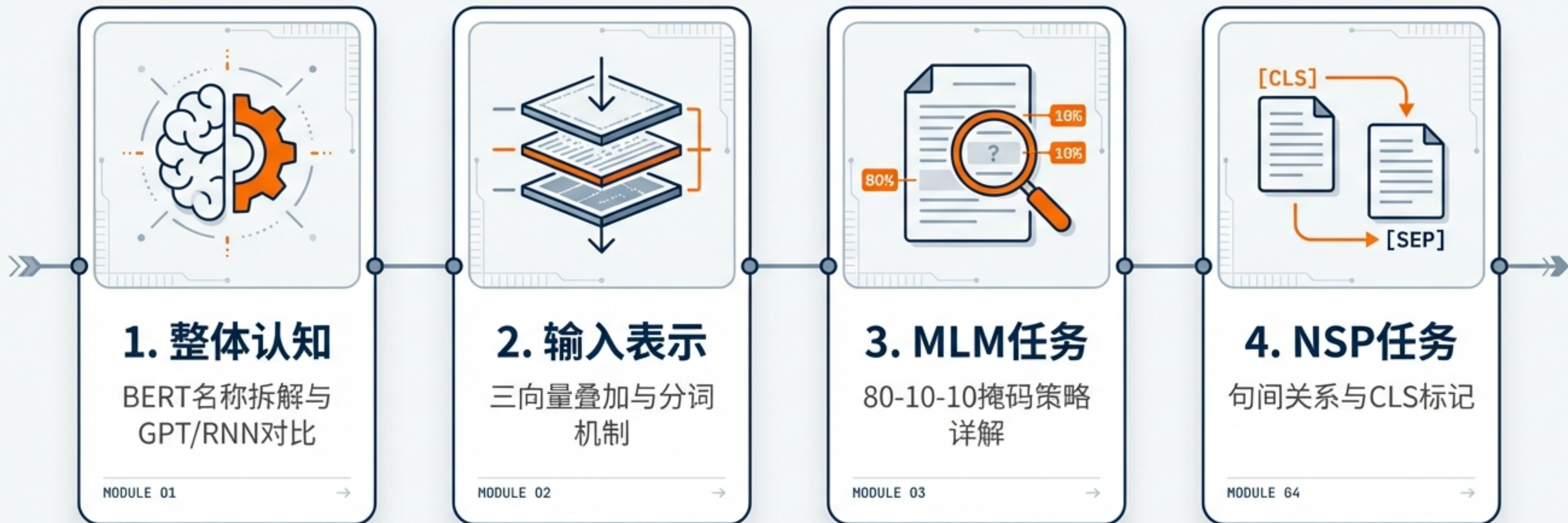


BERT核心原理及预训练机制

像AI专家一样思考：从架构拆解到训练策略

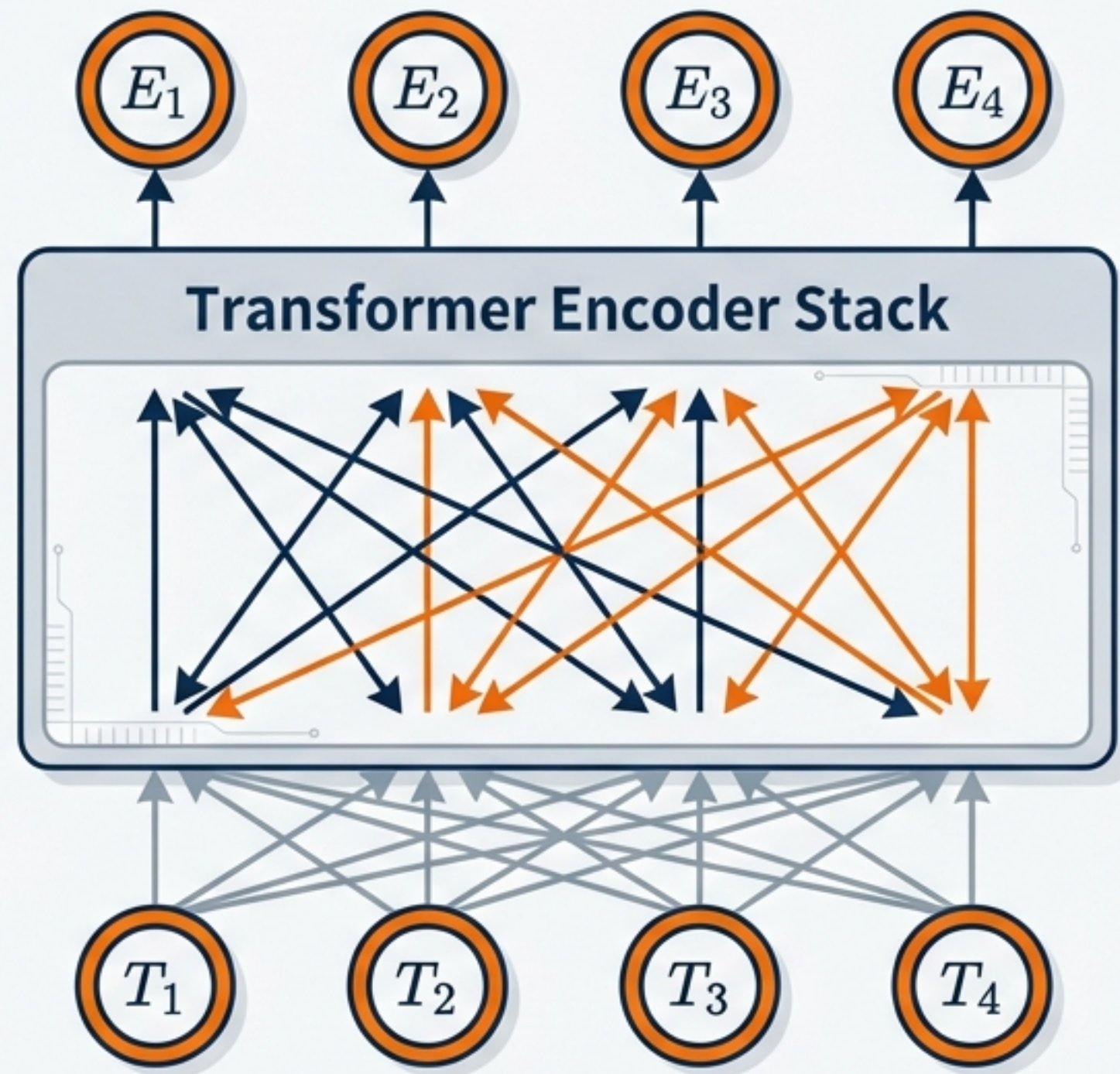


本节课程核心大纲



BERT的本质：双向Transformer编码器

- B** – **Bidirectional** (双向)
同时利用上下文信息
- E** – **Encoder** (编码器)
负责理解和提取特征
- R** – **Representations** (表征)
学习到的上下文相关向量
- T** – **Transformers**
基于自注意力机制的架构



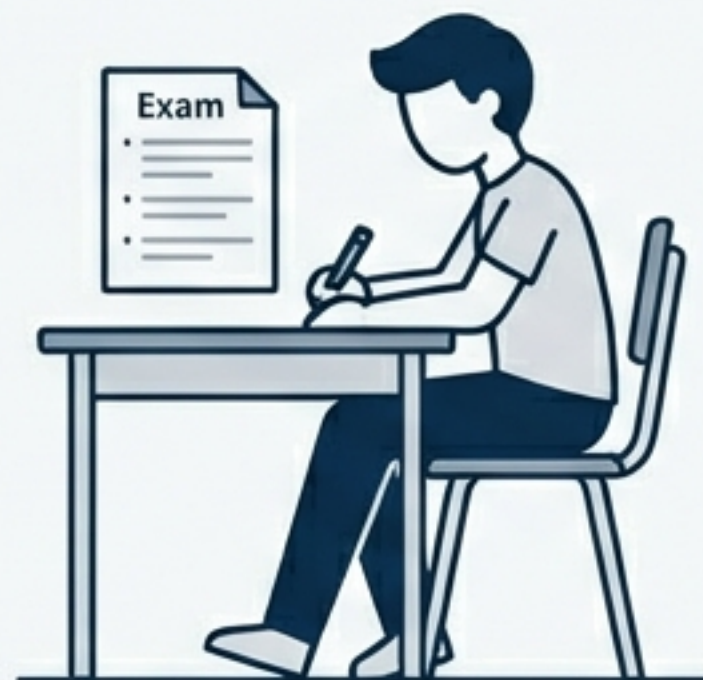
核心训练哲学：预训练 + 微调

Pre-training / 预训练



- 海量无标签文本
- 培养通用“语感”
- 任务：MLM + NSP

Fine-tuning / 微调



- 针对具体下游任务（QA，情感分析）
- 少量数据调整参数
- 任务：特定分类/预测

核心价值：一次预训练，多处通用（Train once, apply everywhere）

进化之路：BERT vs RNN/LSTM

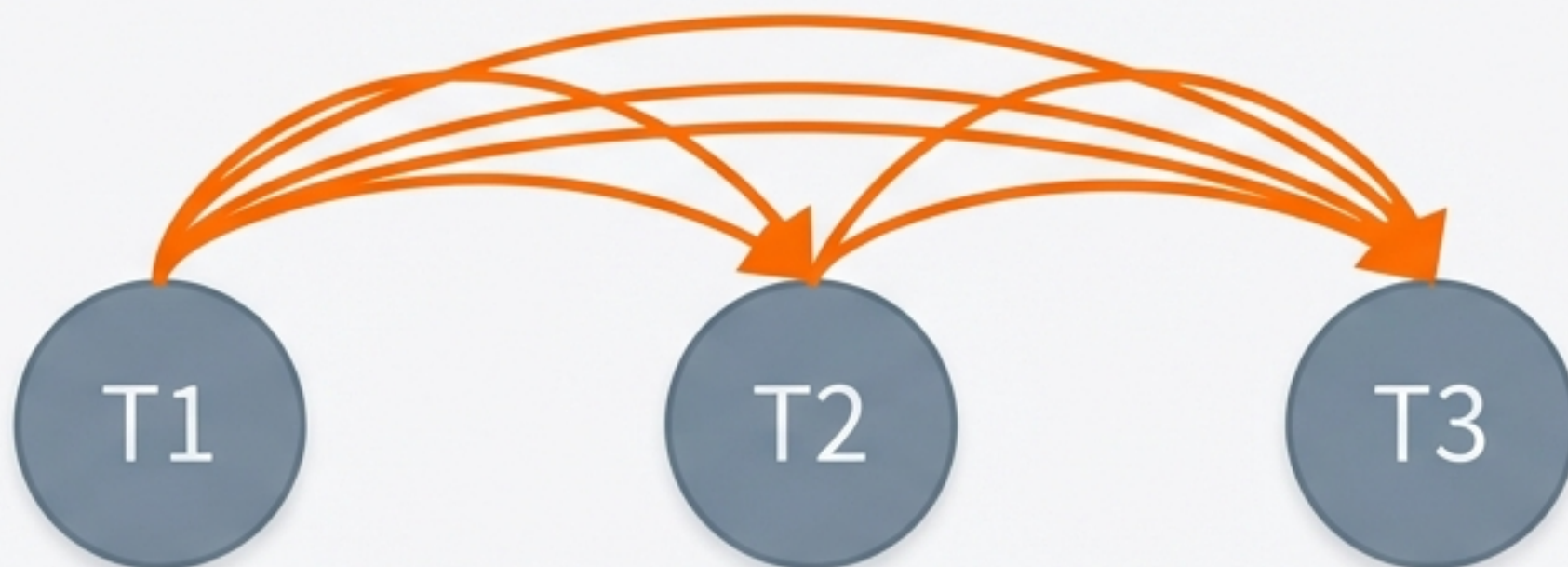
串行结构
(Sequential)



必须逐词阅读，无法并行

长距离依赖丢失

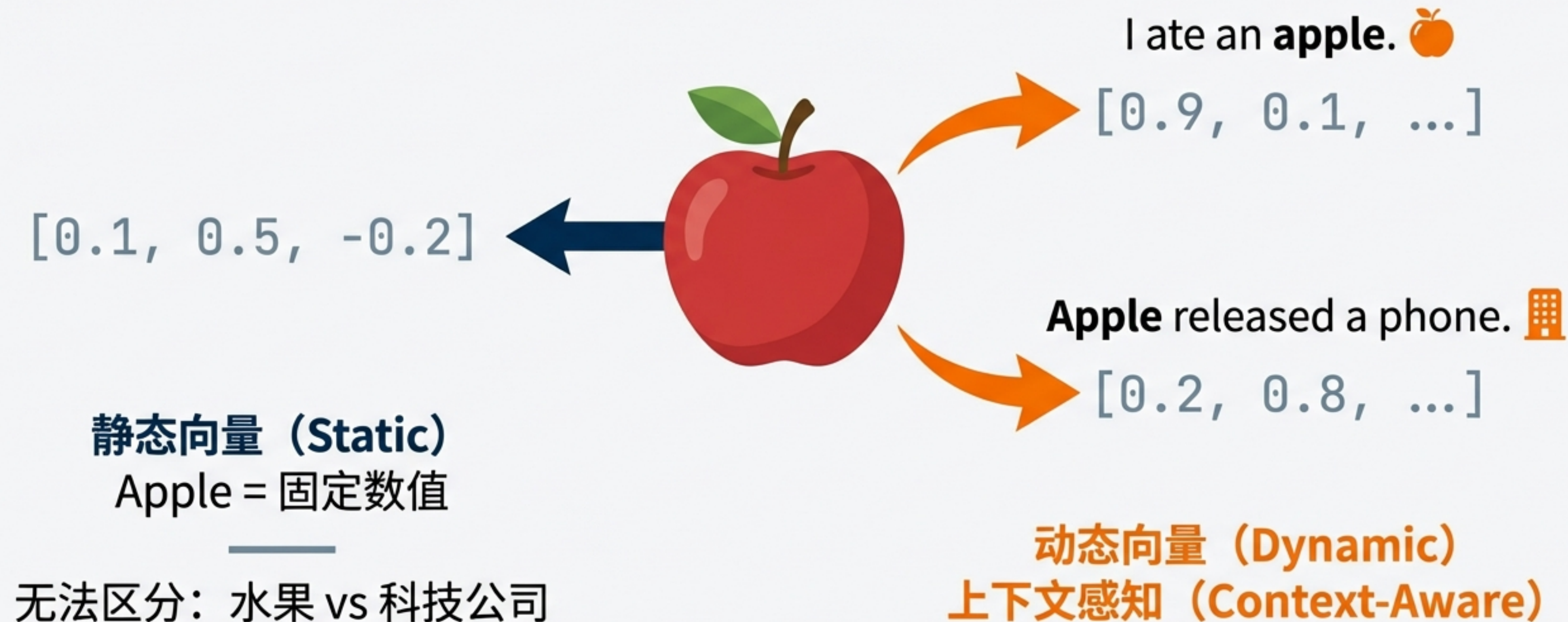
并行结构
(Parallel - Self Attention)



一眼扫完整句话

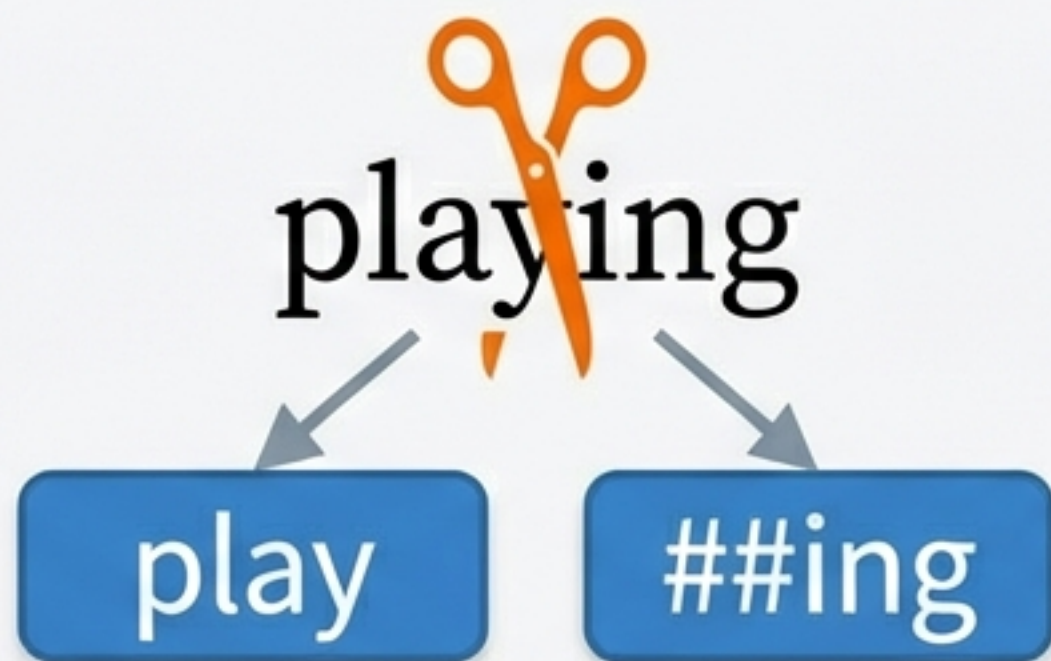
捕捉长距离依赖 (虽然...但是...)

语义理解的飞跃：动态向量 vs 静态向量



预处理第一步：分词机制

English (WordPiece)



处理未登录词 (OOV)
通过词根词缀理解生僻词

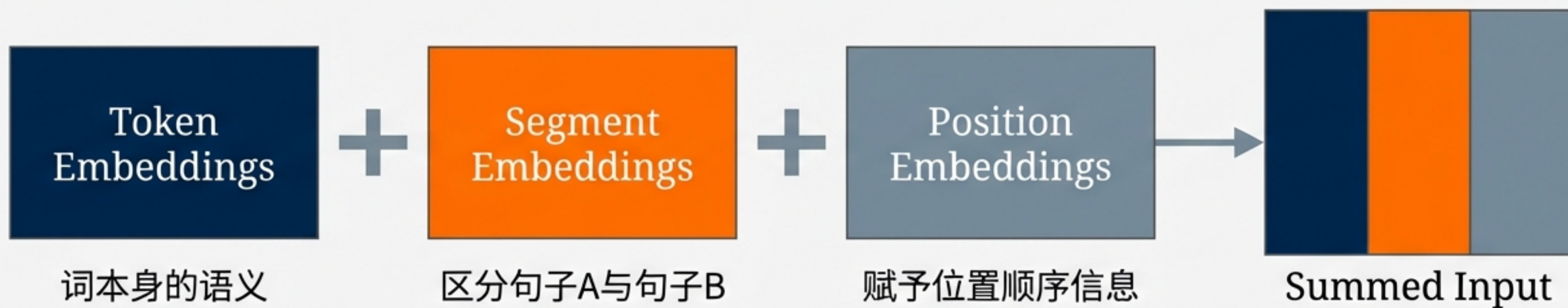
Chinese (Character-based)



按字切分
防止分词错误，覆盖全面

所有Token最终均映射为向量表中的ID

输入表示：三向量叠加模型



$$\text{Input Representation} = \text{Token} + \text{Segment} + \text{Position}$$

实例演示：输入构建五步法

1. Raw Text

Text A: Beijing is a beautiful city | Text B: I love Beijing

2. Tokenization

[CLS] Beijing is a beautiful city [SEP] I love Beijing [SEP]

3. Token Embedding

[CLS] Beijing is a beautiful city [SEP] I love Beijing [SEP]

4. Segment Embedding

[CLS] Beijing is a beautiful city [SEP] I love Beijing [SEP]

5. Position Embedding

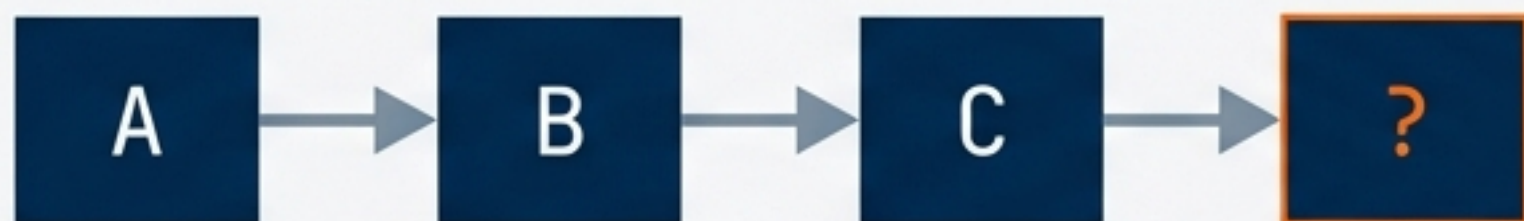
[CLS] Beijing is a beautiful city [SEP] I love Beijing [SEP]

Summation
(Token + Segment + Position)

Σ Encoder Input

MLM任务：双向理解的“完形填空”

GPT (Auto-regressive / 自回归)



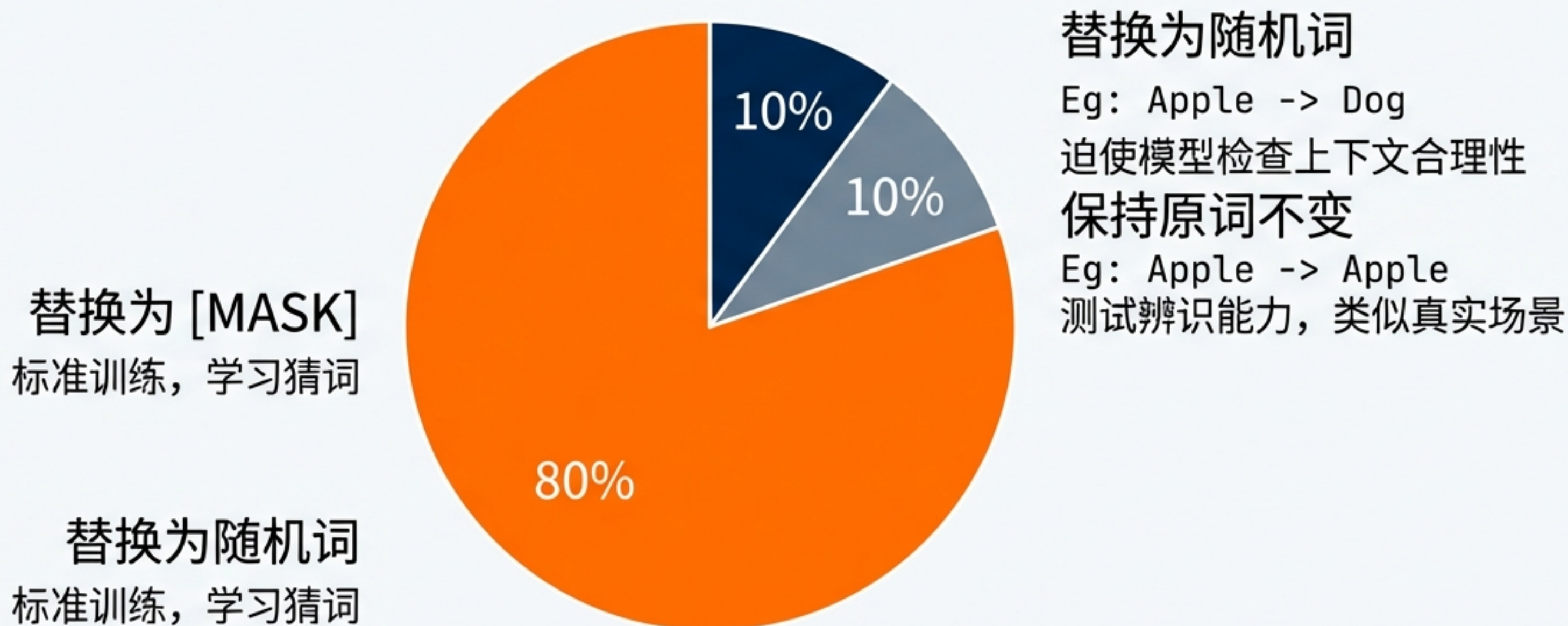
单向：只能看历史

BERT (Auto-encoding / 自编码)



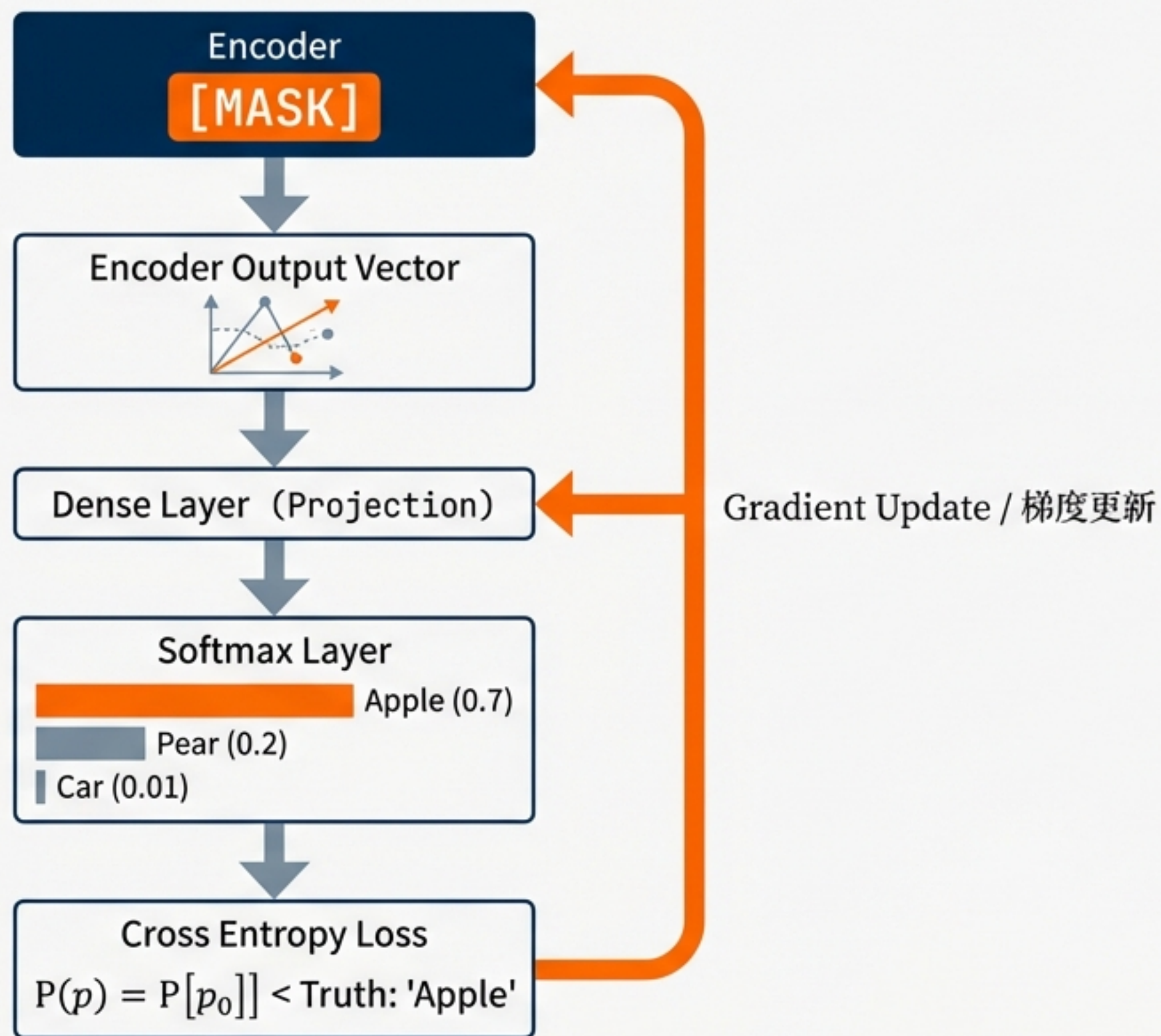
双向：同时利用左右上下文

关键策略：15%掩码与80-10-10法则



目的：减少对 [MASK] 标记的依赖，提升鲁棒性

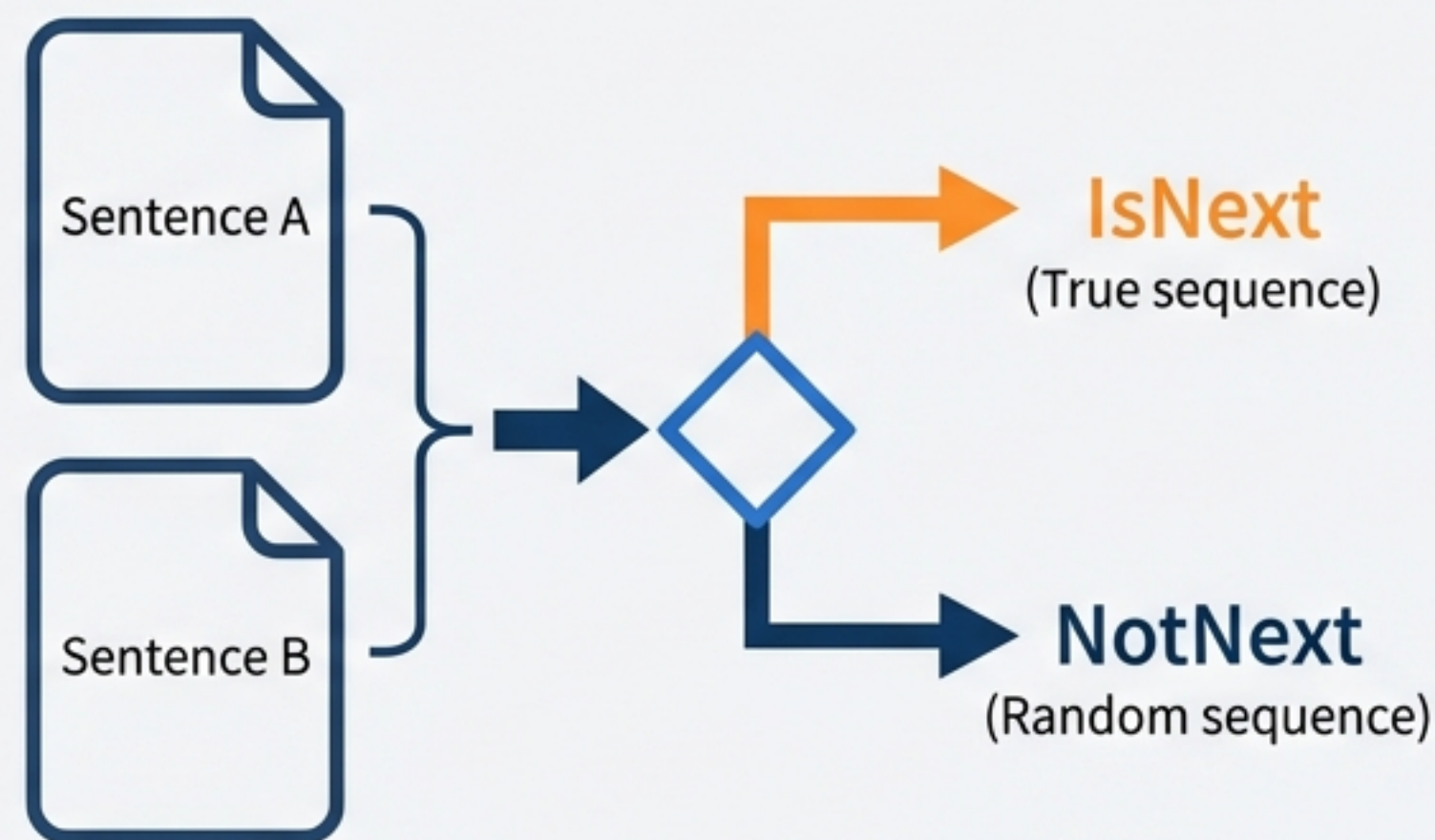
MLM训练流程：从预测到损失计算



NSP任务：理解句子间的逻辑关系

Next Sentence Prediction (下一句预测)

- 类型：二分类任务 (Binary Classification)
- 目标：判断句子B是否紧接句子A



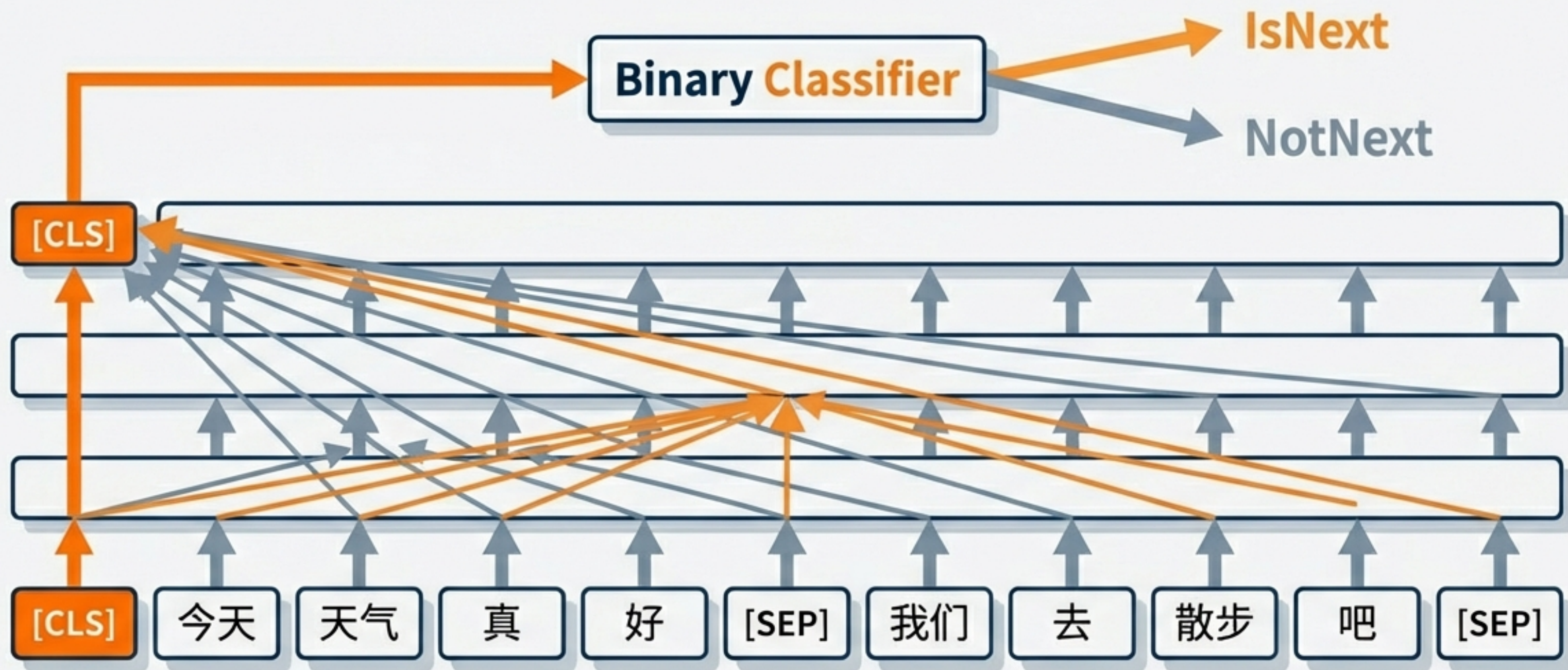
关键应用于问答(QA)与推理任务

NSP数据采样：正负样本均衡



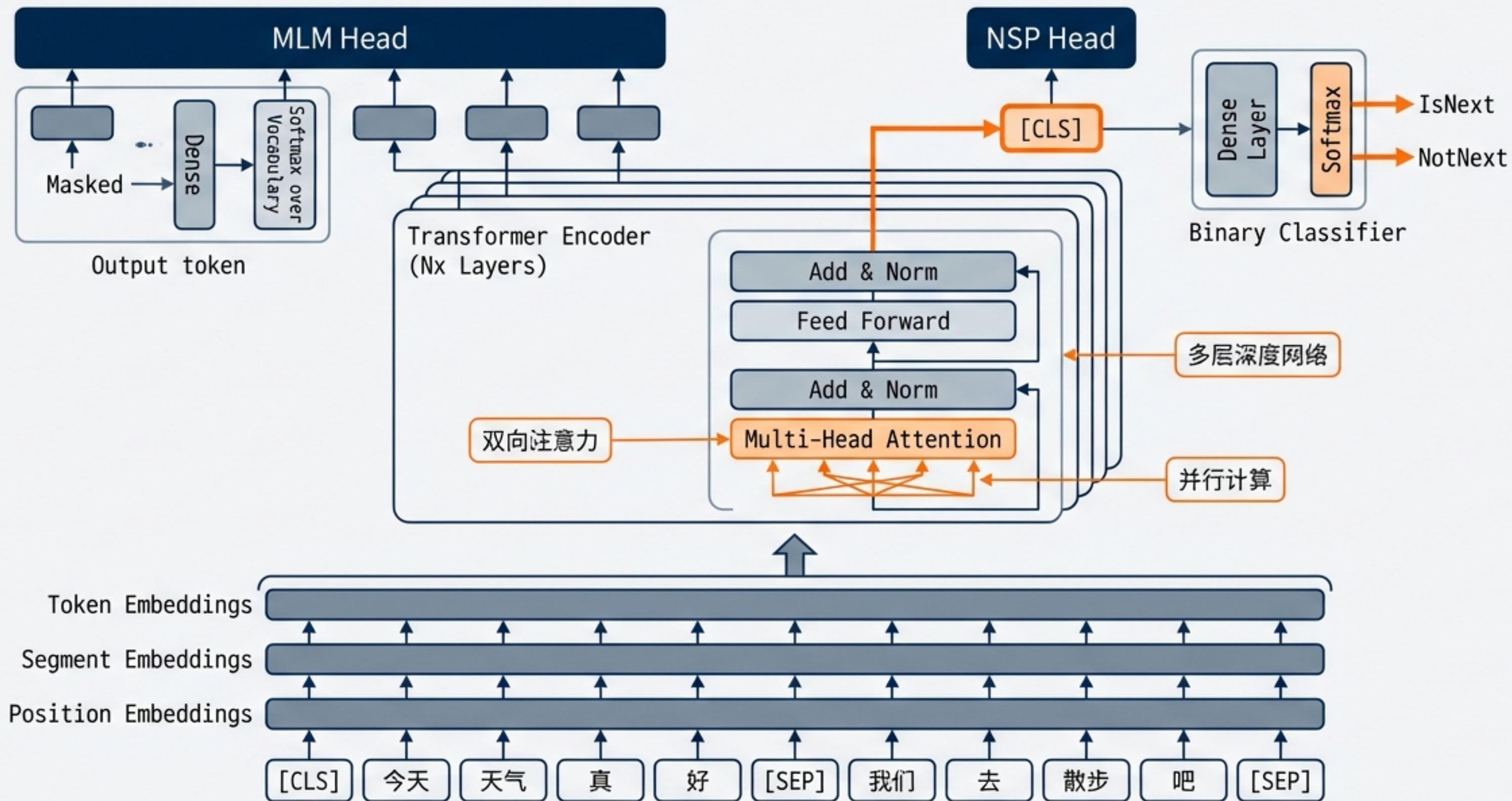
强迫模型学习逻辑连贯性与因果关系

[CLS]标记：全局信息的集大成者

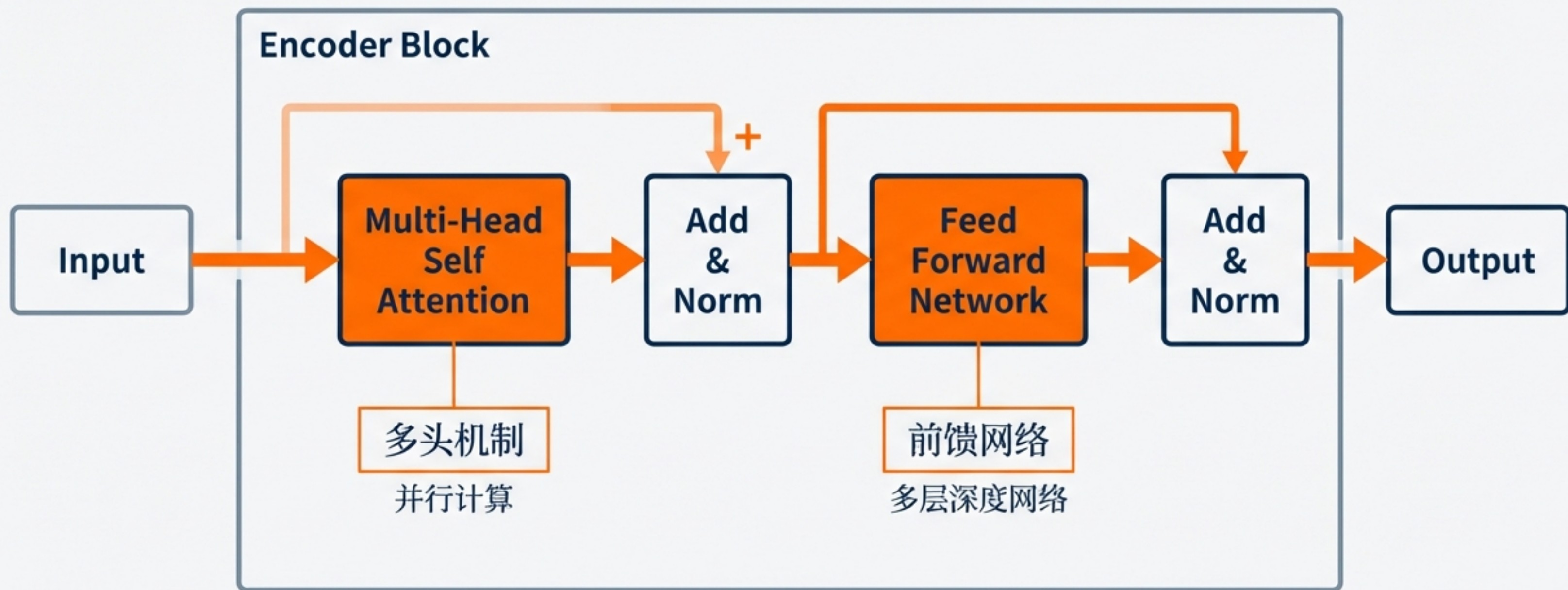


[CLS] 向量聚合了全局信息，作为整句关系的判断依据

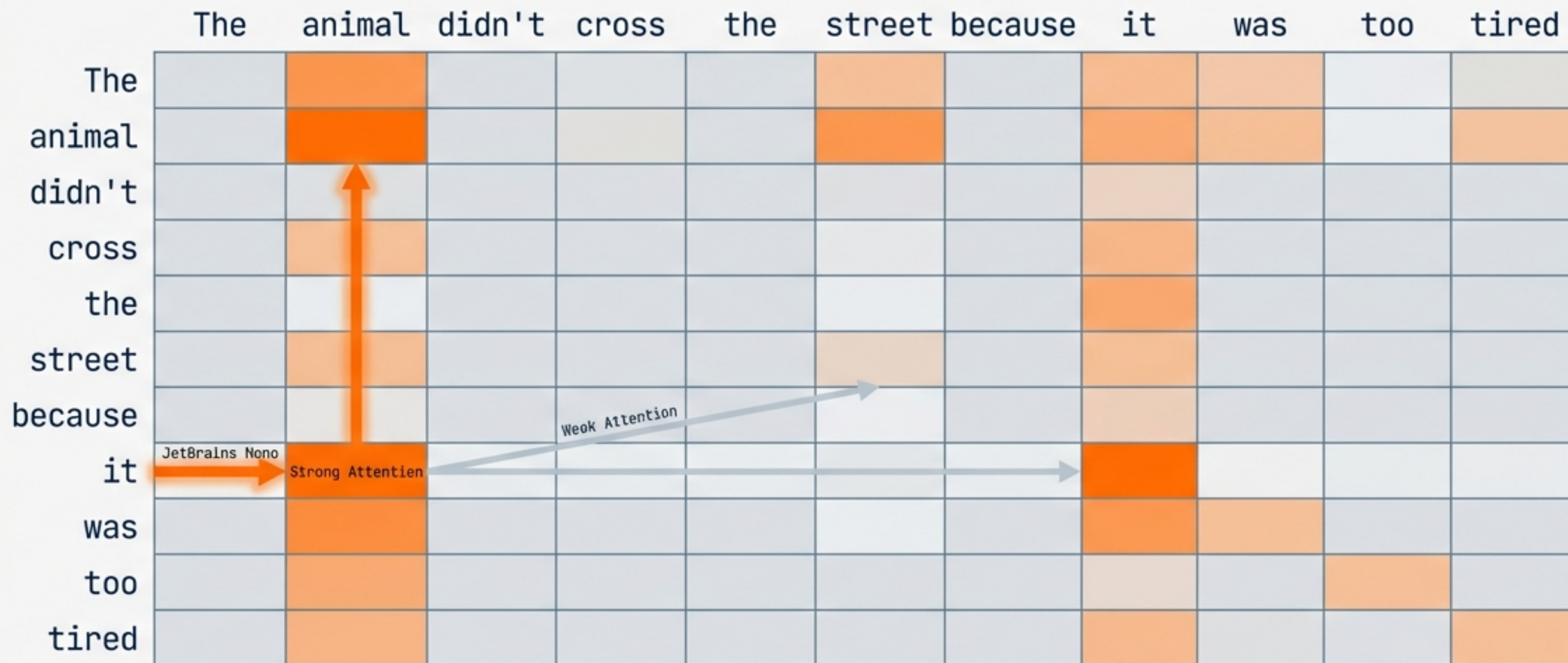
BERT架构总结与全景图



Transformer Encoder 细节详解



注意力机制可视化的意义

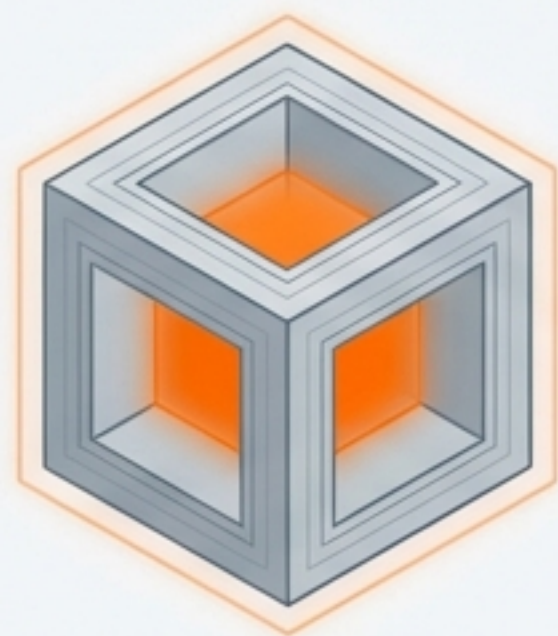


Self-Attention让模型知道“it”指代的是“animal”

BERT模型参数规模

BERT-Base

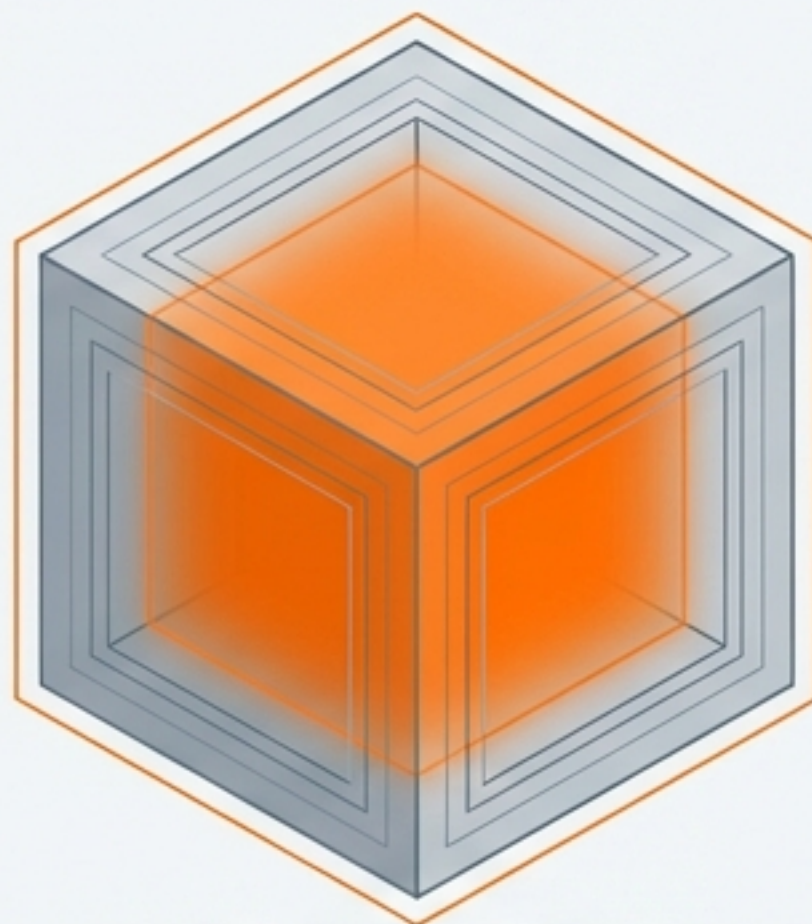
- 12 Layers (层数)
- 768 Hidden Size (隐藏层维度)
- 12 Attention Heads (注意力头数)
- 110M Parameters (参数量)



BERT-Base Volume

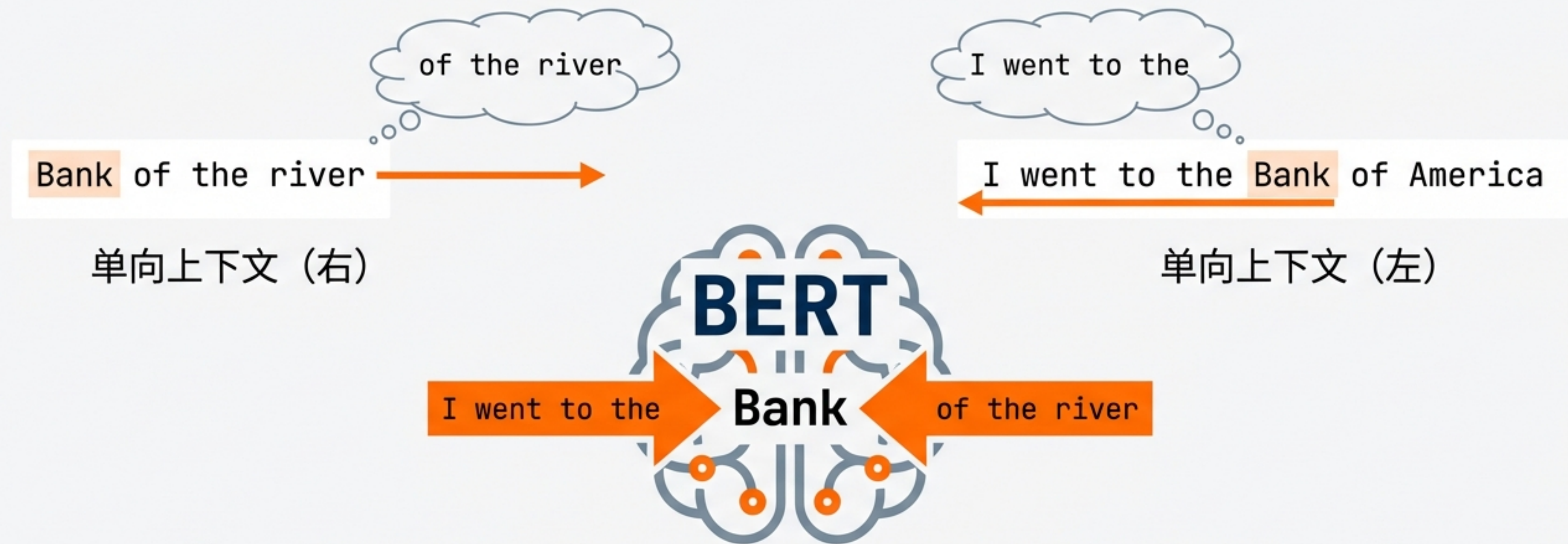
BERT-Large

- 24 Layers
- 1024 Hidden Size
- 16 Attention Heads
- 340M Parameters



BERT-Large Volume

为什么需要双向上下文?



消除歧义 (**Disambiguation**) 需要全向信息流